

Forecasting Competition Challenges

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Github Repo

https://github.com/lingyuehao/Oh_Hao_ENV797_TSA_ForecastCompetition_S26

Setup

```
library(readxl)
library(dplyr)
library(tidyr)
library(lubridate)
library(stringr)
library(forecast)
library(ggplot2)
```

Load Data

```
load_raw <- read_excel("data/load.xlsx")
temp_raw <- read_excel("data/temperature.xlsx")
rh_raw <- read_excel("data/relative_humidity.xlsx")
sub_template <- read_csv("output/submission_template.csv")
```

```
glimpse(load_raw)
```

```
## Rows: 2,372
## Columns: 26
## $ meter_id <chr> "0072", "0072", "0072", "0072", "0072", "0072", "0072", "0072~
## $ date      <dtm> 2005-01-01, 2005-01-02, 2005-01-03, 2005-01-04, 2005-01-05, ~
## $ h1       <dbl> 3072, 2408, 2575, 2177, 1912, 2104, 2336, 2849, 3264, 3520, 2~
## $ h2       <dbl> 2911, 2328, 2528, 2016, 1768, 1976, 2296, 2695, 3199, 3568, 2~
## $ h3       <dbl> 2784, 2311, 2496, 1951, 1697, 1920, 2336, 2640, 3240, 3584, 2~
## $ h4       <dbl> 2768, 2328, 2608, 1913, 1720, 1952, 2448, 2600, 3336, 3728, 3~
## $ h5       <dbl> 2832, 2400, 2825, 2080, 1880, 2167, 2712, 2728, 3480, 3943, 3~
## $ h6       <dbl> 2904, 2552, 3376, 2624, 2472, 2736, 3408, 2984, 3641, 4680, 4~
## $ h7       <dbl> 3064, 2856, 3760, 3328, 3208, 3432, 4248, 3144, 4015, 5256, 4~
```

```
## $ h8      <dbl> 3192, 3272, 3672, 2896, 2888, 3096, 3943, 3432, 4568, 5000, 4~
## $ h9      <dbl> 3425, 4008, 3432, 2600, 2592, 2712, 3504, 3848, 5192, 4416, 3~
## $ h10     <dbl> 3640, 4207, 3440, 2480, 2383, 2672, 3329, 4152, 4903, 3704, 3~
## $ h11     <dbl> 3512, 3752, 3136, 2287, 2160, 2616, 2983, 4072, 4032, 3136, 3~
## $ h12     <dbl> 3408, 2952, 2760, 2177, 2120, 2448, 2736, 3944, 3329, 2840, 3~
## $ h13     <dbl> 3247, 2984, 2640, 2152, 2112, 2360, 2736, 3624, 3144, 2616, 2~
## $ h14     <dbl> 2993, 2920, 2609, 2120, 2008, 2311, 2688, 3280, 3000, 2383, 2~
## $ h15     <dbl> 2839, 2808, 2496, 2128, 1968, 2264, 2705, 2992, 2856, 2216, 2~
## $ h16     <dbl> 2768, 2648, 2424, 2168, 2160, 2384, 2743, 2873, 2776, 2272, 2~
## $ h17     <dbl> 2800, 2856, 2647, 2376, 2376, 2784, 3113, 2904, 3024, 2585, 2~
## $ h18     <dbl> 3200, 3424, 3368, 3072, 3113, 3415, 3688, 3360, 3648, 3319, 3~
## $ h19     <dbl> 3576, 3704, 3824, 3600, 3679, 3737, 4064, 3775, 4328, 4064, 4~
## $ h20     <dbl> 3568, 3784, 4000, 3655, 3600, 3720, 4024, 3833, 4543, 4184, 4~
## $ h21     <dbl> 3312, 3704, 3824, 3569, 3528, 3736, 3784, 3856, 4601, 4080, 4~
## $ h22     <dbl> 3240, 3448, 3520, 3223, 3329, 3456, 3776, 3800, 4416, 3943, 3~
## $ h23     <dbl> 2904, 3184, 3008, 2736, 2832, 3016, 3552, 3751, 4087, 3384, 3~
## $ h24     <dbl> 2624, 2801, 2503, 2249, 2392, 2624, 3160, 3521, 3713, 3000, 3~
```

```
glimpse(temp_raw)
```

```
## Rows: 56,922
## Columns: 30
## $ date    <dtm> 2005-01-01, 2005-01-01, 2005-01-01, 2005-01-01, 2005-01-01, 20~
## $ hr      <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, ~
## $ t_ws1   <dbl> 43, 41, 40, 39, 38, 37, 37, 36, 36, 44, 51, 56, 61, 64, 66, 66,~
## $ t_ws2   <dbl> 46, 46, 46, 46, 46, 45, 45, 43, 52, 55, 61, 66, 66, 66, 72, 66,~
## $ t_ws3   <dbl> 40, 38, 37, 37, 37, 36, 37, 37, 38, 43, 54, 60, 64, 66, 66, 65,~
## $ t_ws4   <dbl> 47, 46, 45, 47, 44, 45, 45, 43, 49, 55, 59, 63, 67, 67, 68, 68,~
## $ t_ws5   <dbl> 48, 48, 45, 48, 48, 48, 48, 48, 48, 55, 61, 70, 70, 70, 70, 70,~
## $ t_ws6   <dbl> 46, 45, 45, 48, 49, 48, 44, 48, 51, 55, 59, 62, 66, 66, 64, 61,~
## $ t_ws7   <dbl> 44, 51, 49, 45, 43, 40, 41, 43, 51, 56, 62, 64, 66, 68, 69, 68,~
## $ t_ws8   <dbl> 52, 50, 48, 50, 50, 50, 46, 46, 52, 59, 64, 66, 68, 70, 70, 70,~
## $ t_ws9   <dbl> 45, 42, 43, 42, 42, 40, 41, 40, 43, 51, 62, 63, 63, 64, 64, 63,~
## $ t_ws10  <dbl> 42, 41, 40, 39, 38, 37, 37, 36, 35, 45, 52, 58, 63, 67, 69, 70,~
## $ t_ws11  <dbl> 48, 47, 48, 47, 46, 45, 46, 45, 49, 57, 61, 66, 69, 70, 70, 70,~
## $ t_ws12  <dbl> 49, 49, 48, 47, 49, 47, 49, 48, 50, 53, 61, 63, 66, 68, 68, 68,~
## $ t_ws13  <dbl> 48, 48, 41, 46, 46, 43, 43, 45, 48, 54, 55, 57, 63, 66, 68, 68,~
## $ t_ws14  <dbl> 44, 45, 43, 46, 44, 41, 43, 43, 46, 51, 57, 64, 69, 69, 70, 69,~
## $ t_ws15  <dbl> 45, 45, 46, 45, 46, 45, 45, 46, 50, 56, 63, 68, 68, 69, 69, 70,~
## $ t_ws16  <dbl> 53, 53, 52, 52, 51, 50, 50, 51, 54, 55, 56, 57, 57, 56, 56, 56,~
## $ t_ws17  <dbl> 47, 47, 47, 47, 46, 47, 48, 45, 50, 55, 63, 67, 71, 73, 72, 71,~
## $ t_ws18  <dbl> 50, 48, 46, 46, 45, 45, 45, 46, 49, 55, 57, 64, 68, 70, 69, 67,~
## $ t_ws19  <dbl> 39, 41, 43, 43, 43, 39, 39, 39, 43, 54, 55, 63, 66, 66, 66, 66,~
## $ t_ws20  <dbl> 50, 49, 50, 49, 49, 45, 44, 51, 53, 58, 61, 64, 63, 61, 62, 60,~
## $ t_ws21  <dbl> 43, 43, 41, 41, 39, 39, 39, 39, 43, 52, 57, 62, 66, 70, 72, 72,~
## $ t_ws22  <dbl> 45, 46, 45, 43, 43, 46, 45, 45, 52, 57, 66, 70, 70, 70, 70, 70,~
## $ t_ws23  <dbl> 50, 47, 45, 45, 43, 41, 44, 47, 51, 56, 61, 63, 67, 68, 70, 69,~
## $ t_ws24  <dbl> 47, 47, 46, 44, 45, 48, 47, 49, 51, 55, 59, 63, 65, 66, 68, 65,~
## $ t_ws25  <dbl> 41, 45, 45, 45, 45, 46, 45, 46, 54, 55, 59, 64, 70, 72, 73, 72,~
## $ t_ws26  <dbl> 47, 47, 46, 47, 47, 44, 47, 45, 48, 53, 58, 66, 68, 69, 68, 67,~
## $ t_ws27  <dbl> 47, 47, 46, 47, 46, 46, 47, 47, 48, 54, 61, 65, 69, 72, 71, 72,~
## $ t_ws28  <dbl> 45, 45, 43, 44, 44, 42, 42, 41, 48, 53, 57, 62, 65, 68, 69, 69,~
```

```
glimpse(rh_raw)
```

```
## Rows: 56,921
## Columns: 30
## $ date      <dtm> 2005-01-01, 2005-01-01, 2005-01-01, 2005-01-01, 2005-01-01, 2~
## $ hr        <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, ~
## $ rh_ws1    <dbl> 99, 76, 79, 79, 79, 82, 83, 83, 79, 74, 90, 75, 67, 61, 56, 56~
## $ rh_ws2    <dbl> 93, 93, 93, 93, 93, 93, 93, 93, 88, 82, 63, 64, 64, 60, 57, 68~
## $ rh_ws3    <dbl> 93, 97, 93, 93, 96, 97, 90, 88, 93, 89, 69, 53, 46, 43, 43, 47~
## $ rh_ws4    <dbl> 90, 89, 90, 90, 93, 97, 93, 97, 93, 74, 67, 58, 53, 51, 49, 51~
## $ rh_ws5    <dbl> 87, 87, 93, 87, 87, 87, 87, 87, 87, 77, 63, 53, 50, 46, 46, 46~
## $ rh_ws6    <dbl> 93, 97, 97, 89, 90, 93, 100, 93, 89, 80, 75, 67, 65, 68, 70, 7~
## $ rh_ws7    <dbl> 93, 80, 83, 90, 93, 97, 96, 89, 80, 70, 60, 58, 54, 51, 51, 51~
## $ rh_ws8    <dbl> 82, 88, 87, 82, 82, 88, 93, 93, 82, 72, 64, 60, 53, 50, 50, 50~
## $ rh_ws9    <dbl> 67, 67, 63, 63, 59, 63, 63, 67, 67, 55, 56, 52, 56, 52, 52, 52~
## $ rh_ws10   <dbl> 90, 100, 76, 76, 76, 79, 79, 79, 79, 73, 87, 70, 59, 55, 51, 4~
## $ rh_ws11   <dbl> 86, 90, 89, 90, 93, 93, 93, 97, 90, 69, 60, 52, 49, 49, 49, 49~
## $ rh_ws12   <dbl> 93, 93, 96, 100, 97, 100, 97, 96, 93, 89, 70, 68, 61, 57, 55, ~
## $ rh_ws13   <dbl> 71, 66, 70, 71, 66, 66, 71, 71, 66, 67, 67, 67, 68, 68, 69, 69~
## $ rh_ws14   <dbl> 93, 90, 93, 89, 89, 96, 93, 89, 86, 77, 64, 52, 45, 45, 44, 45~
## $ rh_ws15   <dbl> 87, 93, 87, 93, 87, 93, 93, 87, 88, 74, 60, 53, 52, 55, 54, 50~
## $ rh_ws16   <dbl> 77, 74, 77, 77, 80, 80, 80, 80, 72, 69, 67, 67, 69, 72, 72, 72~
## $ rh_ws17   <dbl> 86, 86, 86, 86, 89, 90, 86, 93, 83, 72, 56, 49, 46, 43, 46, 47~
## $ rh_ws18   <dbl> 89, 93, 93, 93, 90, 90, 93, 89, 93, 77, 72, 58, 51, 47, 51, 55~
## $ rh_ws19   <dbl> 93, 87, 87, 87, 87, 93, 93, 93, 87, 82, 72, 56, 49, 49, 49, 49~
## $ rh_ws20   <dbl> 93, 97, 96, 97, 97, 100, 100, 83, 86, 72, 67, 61, 68, 70, 67, ~
## $ rh_ws21   <dbl> 93, 87, 93, 93, 93, 93, 93, 100, 93, 82, 63, 54, 43, 40, 38, 4~
## $ rh_ws22   <dbl> 81, 82, 81, 81, 87, 82, 81, 81, 72, 63, 49, 40, 43, 43, 40, 43~
## $ rh_ws23   <dbl> 83, 86, 93, 93, 97, 96, 93, 93, 86, 75, 65, 63, 53, 53, 49, 53~
## $ rh_ws24   <dbl> 97, 93, 96, 100, 97, 93, 97, 93, 89, 80, 72, 68, 66, 65, 61, 7~
## $ rh_ws25   <dbl> 87, 81, 76, 81, 81, 76, 81, 82, 72, 67, 63, 52, 43, 41, 41, 44~
## $ rh_ws26   <dbl> 83, 83, 86, 86, 86, 89, 86, 90, 86, 77, 67, 54, 53, 51, 65, 65~
## $ rh_ws27   <dbl> 86, 86, 89, 90, 89, 89, 90, 90, 89, 77, 60, 52, 47, 44, 46, 46~
## $ rh_ws28   <dbl> 86, 86, 93, 93, 93, 96, 96, 96, 93, 80, 72, 65, 59, 55, 53, 53~
```

```
names(load_raw)
```

```
## [1] "meter_id" "date"      "h1"      "h2"      "h3"      "h4"
## [7] "h5"       "h6"       "h7"       "h8"       "h9"       "h10"
## [13] "h11"      "h12"      "h13"      "h14"      "h15"      "h16"
## [19] "h17"      "h18"      "h19"      "h20"      "h21"      "h22"
## [25] "h23"      "h24"
```

```
names(temp_raw)
```

```
## [1] "date"      "hr"      "t_ws1"   "t_ws2"   "t_ws3"   "t_ws4"   "t_ws5"   "t_ws6"
## [9] "t_ws7"     "t_ws8"   "t_ws9"   "t_ws10"  "t_ws11"  "t_ws12"  "t_ws13"  "t_ws14"
## [17] "t_ws15"    "t_ws16"  "t_ws17"  "t_ws18"  "t_ws19"  "t_ws20"  "t_ws21"  "t_ws22"
## [25] "t_ws23"    "t_ws24"  "t_ws25"  "t_ws26"  "t_ws27"  "t_ws28"
```

```
names(rh_raw)
```

```
## [1] "date"      "hr"        "rh_ws1"    "rh_ws2"    "rh_ws3"    "rh_ws4"    "rh_ws5"
## [8] "rh_ws6"    "rh_ws7"    "rh_ws8"    "rh_ws9"    "rh_ws10"   "rh_ws11"   "rh_ws12"
## [15] "rh_ws13"   "rh_ws14"   "rh_ws15"   "rh_ws16"   "rh_ws17"   "rh_ws18"   "rh_ws19"
## [22] "rh_ws20"   "rh_ws21"   "rh_ws22"   "rh_ws23"   "rh_ws24"   "rh_ws25"   "rh_ws26"
## [29] "rh_ws27"   "rh_ws28"
```

```
head(load_raw)
```

```
## # A tibble: 6 x 26
##   meter_id date                h1    h2    h3    h4    h5    h6    h7    h8
##   <chr>      <dtm>              <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 0072      2005-01-01 00:00:00 3072 2911 2784 2768 2832 2904 3064 3192
## 2 0072      2005-01-02 00:00:00 2408 2328 2311 2328 2400 2552 2856 3272
## 3 0072      2005-01-03 00:00:00 2575 2528 2496 2608 2825 3376 3760 3672
## 4 0072      2005-01-04 00:00:00 2177 2016 1951 1913 2080 2624 3328 2896
## 5 0072      2005-01-05 00:00:00 1912 1768 1697 1720 1880 2472 3208 2888
## 6 0072      2005-01-06 00:00:00 2104 1976 1920 1952 2167 2736 3432 3096
## # i 16 more variables: h9 <dbl>, h10 <dbl>, h11 <dbl>, h12 <dbl>, h13 <dbl>,
## #   h14 <dbl>, h15 <dbl>, h16 <dbl>, h17 <dbl>, h18 <dbl>, h19 <dbl>,
## #   h20 <dbl>, h21 <dbl>, h22 <dbl>, h23 <dbl>, h24 <dbl>
```

```
head(temp_raw)
```

```
## # A tibble: 6 x 30
##   date                hr t_ws1 t_ws2 t_ws3 t_ws4 t_ws5 t_ws6 t_ws7 t_ws8
##   <dtm>              <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 2005-01-01 00:00:00    1   43   46   40   47   48   46   44   52
## 2 2005-01-01 00:00:00    2   41   46   38   46   48   45   51   50
## 3 2005-01-01 00:00:00    3   40   46   37   45   45   45   49   48
## 4 2005-01-01 00:00:00    4   39   46   37   47   48   48   45   50
## 5 2005-01-01 00:00:00    5   38   46   37   44   48   49   43   50
## 6 2005-01-01 00:00:00    6   37   45   36   45   48   48   40   50
## # i 20 more variables: t_ws9 <dbl>, t_ws10 <dbl>, t_ws11 <dbl>, t_ws12 <dbl>,
## #   t_ws13 <dbl>, t_ws14 <dbl>, t_ws15 <dbl>, t_ws16 <dbl>, t_ws17 <dbl>,
## #   t_ws18 <dbl>, t_ws19 <dbl>, t_ws20 <dbl>, t_ws21 <dbl>, t_ws22 <dbl>,
## #   t_ws23 <dbl>, t_ws24 <dbl>, t_ws25 <dbl>, t_ws26 <dbl>, t_ws27 <dbl>,
## #   t_ws28 <dbl>
```

```
head(rh_raw)
```

```
## # A tibble: 6 x 30
##   date                hr rh_ws1 rh_ws2 rh_ws3 rh_ws4 rh_ws5 rh_ws6 rh_ws7
##   <dtm>              <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 2005-01-01 00:00:00    1   99   93   93   90   87   93   93
## 2 2005-01-01 00:00:00    2   76   93   97   89   87   97   80
## 3 2005-01-01 00:00:00    3   79   93   93   90   93   97   83
## 4 2005-01-01 00:00:00    4   79   93   93   90   87   89   90
## 5 2005-01-01 00:00:00    5   79   93   96   93   87   90   93
## 6 2005-01-01 00:00:00    6   82   93   97   97   87   93   97
```

```
## # i 21 more variables: rh_ws8 <dbl>, rh_ws9 <dbl>, rh_ws10 <dbl>,
## #   rh_ws11 <dbl>, rh_ws12 <dbl>, rh_ws13 <dbl>, rh_ws14 <dbl>, rh_ws15 <dbl>,
## #   rh_ws16 <dbl>, rh_ws17 <dbl>, rh_ws18 <dbl>, rh_ws19 <dbl>, rh_ws20 <dbl>,
## #   rh_ws21 <dbl>, rh_ws22 <dbl>, rh_ws23 <dbl>, rh_ws24 <dbl>, rh_ws25 <dbl>,
## #   rh_ws26 <dbl>, rh_ws27 <dbl>, rh_ws28 <dbl>
```

```
tail(load_raw$date)
```

```
## [1] "2011-06-25 UTC" "2011-06-26 UTC" "2011-06-27 UTC" "2011-06-28 UTC"
## [5] "2011-06-29 UTC" "2011-06-30 UTC"
```

```
tail(temp_raw$date)
```

```
## [1] "2011-06-30 UTC" "2011-06-30 UTC" "2011-06-30 UTC" "2011-06-30 UTC"
## [5] "2011-06-30 UTC" NA
```

```
tail(rh_raw$date)
```

```
## [1] "2011-06-30 UTC" "2011-06-30 UTC" "2011-06-30 UTC" "2011-06-30 UTC"
## [5] "2011-06-30 UTC" "2011-06-30 UTC"
```

Data Preprocessing

```
## transfer load to daily
load_daily <- load_raw %>%
  mutate(date = as.Date(date)) %>%
  mutate(
    load_daily = rowMeans(select(., starts_with("h")), na.rm = TRUE)
  ) %>%
  select(date, load_daily)
```

```
## transfer temp to daily
temp_daily <- temp_raw %>%
  mutate(date = as.Date(date)) %>%
  mutate(
    temp_hourly_avg = rowMeans(select(., starts_with("t_ws")), na.rm = TRUE)
  ) %>%
  group_by(date) %>%
  summarise(
    temp_daily = mean(temp_hourly_avg, na.rm = TRUE),
    .groups = "drop"
  )
```

```
## transfer humidity to daily
rh_daily <- rh_raw %>%
  mutate(date = as.Date(date)) %>%
  mutate(
    rh_hourly_avg = rowMeans(select(., starts_with("rh_ws")), na.rm = TRUE)
```

```
) %>%
group_by(date) %>%
summarise(
  rh_daily = mean(rh_hourly_avg, na.rm = TRUE),
  .groups = "drop"
)
```

```
## combine daily data
daily_data <- load_daily %>%
  left_join(temp_daily, by = "date") %>%
  left_join(rh_daily, by = "date")

glimpse(daily_data)
```

```
## Rows: 2,372
## Columns: 4
## $ date      <date> 2005-01-01, 2005-01-02, 2005-01-03, 2005-01-04, 2005-01-05~
## $ load_daily <dbl> 3107.625, 3068.292, 3061.292, 2565.708, 2495.708, 2734.917,~
## $ temp_daily <dbl> 53.56548, 53.75744, 55.91369, 61.69494, 60.43006, 62.00446,~
## $ rh_daily   <dbl> 76.70982, 80.47917, 81.22917, 74.84226, 76.06696, 77.99107,~
```

```
head(daily_data)
```

```
## # A tibble: 6 x 4
##   date      load_daily temp_daily rh_daily
##   <date>      <dbl>      <dbl>   <dbl>
## 1 2005-01-01    3108.        53.6    76.7
## 2 2005-01-02    3068.        53.8    80.5
## 3 2005-01-03    3061.        55.9    81.2
## 4 2005-01-04    2566.        61.7    74.8
## 5 2005-01-05    2496.        60.4    76.1
## 6 2005-01-06    2735.        62.0    78.0
```

```
summary(daily_data)
```

```
##      date      load_daily      temp_daily      rh_daily
## Min.   :2005-01-01 Min.   :2247 Min.   :21.39 Min.   :27.10
## 1st Qu.:2006-08-16 1st Qu.:2798 1st Qu.:48.53 1st Qu.:61.12
## Median :2008-03-31 Median :3506 Median :61.90 Median :69.83
## Mean   :2008-03-31 Mean   :3629 Mean   :60.44 Mean   :68.56
## 3rd Qu.:2009-11-14 3rd Qu.:4211 3rd Qu.:74.12 3rd Qu.:77.01
## Max.   :2011-06-30 Max.   :7897 Max.   :87.09 Max.   :94.95
```

```
colSums(is.na(daily_data))
```

```
##      date load_daily temp_daily rh_daily
##      0          0          0          0
```

```

# load
daily_load <- daily_data %>%
  select(date, load_daily)

# load + temp
daily_load_temp <- daily_data %>%
  select(date, load_daily, temp_daily)

glimpse(daily_load_temp)

## Rows: 2,372
## Columns: 3
## $ date      <date> 2005-01-01, 2005-01-02, 2005-01-03, 2005-01-04, 2005-01-05~
## $ load_daily <dbl> 3107.625, 3068.292, 3061.292, 2565.708, 2495.708, 2734.917,~
## $ temp_daily <dbl> 53.56548, 53.75744, 55.91369, 61.69494, 60.43006, 62.00446,~

```

Train/Validation split

```

# split train and validation data (load + temp + humid)
train_data <- daily_data %>%
  filter(date <= as.Date("2010-06-30"))

valid_data <- daily_data %>%
  filter(date >= as.Date("2010-07-01") &
         date <= as.Date("2010-07-31"))

# split train and validation data (load)
train_data_load <- daily_load %>%
  filter(date <= as.Date("2010-06-30"))

valid_data_load <- daily_load %>%
  filter(date >= as.Date("2010-07-01") &
         date <= as.Date("2010-07-31"))

# split train and validation data (load + temp)
train_data_load_temp <- daily_load_temp %>%
  filter(date <= as.Date("2010-06-30"))

valid_data_load_temp <- daily_load_temp %>%
  filter(date >= as.Date("2010-07-01") &
         date <= as.Date("2010-07-31"))

# use all available historical data up to 2011-06-30
full_data <- daily_data %>%
  filter(date >= as.Date("2005-01-01"),
         date <= as.Date("2011-06-30")) %>%
  arrange(date)

full_data_load <- daily_load %>%
  filter(date >= as.Date("2005-01-01"),

```

```

        date <= as.Date("2011-06-30")) %>%
    arrange(date)

full_data_load_temp <- daily_load_temp %>%
    filter(date >= as.Date("2005-01-01"),
           date <= as.Date("2011-06-30")) %>%
    arrange(date)

# create time series with weekly seasonality
train_ts <- ts(train_data$load_daily, frequency = 7)
train_msts <- msts(train_data$load_daily, seasonal.periods = c(7, 365))

train_ts_load <- ts(train_data_load$load_daily, frequency = 7)
train_msts_load <- msts(train_data_load$load_daily, seasonal.periods = c(7, 365))

train_ts_load_temp <- ts(train_data_load_temp$load_daily, frequency = 7)
train_msts_load_temp <- msts(train_data_load_temp$load_daily, seasonal.periods = c(7, 365))

# create time series with weekly seasonality
full_ts <- ts(full_data$load_daily, frequency = 7)
full_msts <- msts(full_data$load_daily, seasonal.periods = c(7, 365))

full_ts_load <- ts(full_data_load$load_daily, frequency = 7)
full_msts_load <- msts(full_data_load$load_daily, seasonal.periods = c(7, 365))

full_ts_load_temp <- ts(full_data_load_temp$load_daily, frequency = 7)
full_msts_load_temp <- msts(full_data_load_temp$load_daily, seasonal.periods = c(7, 365))

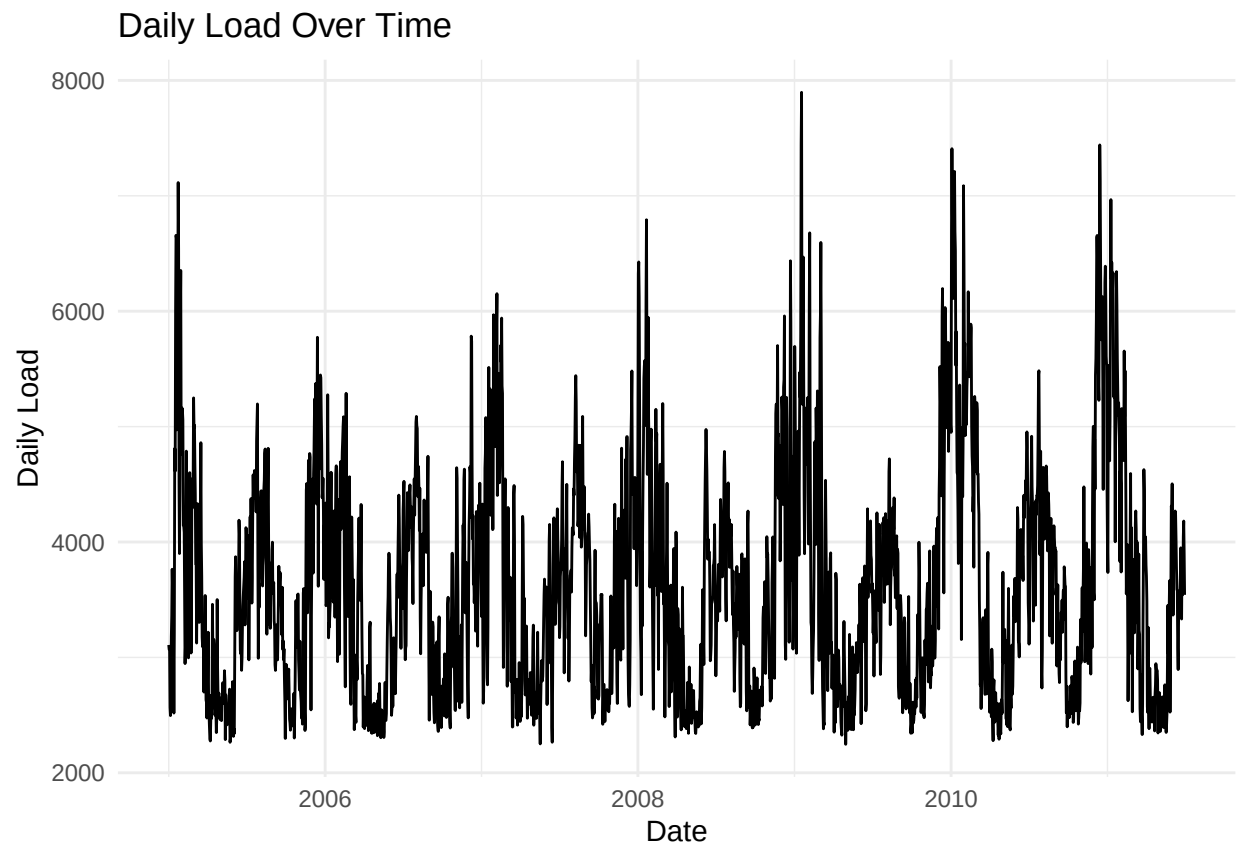
```

EDA

```

ggplot(daily_load, aes(x = date, y = load_daily)) +
  geom_line() +
  labs(title = "Daily Load Over Time", x = "Date", y = "Daily Load") +
  theme_minimal()

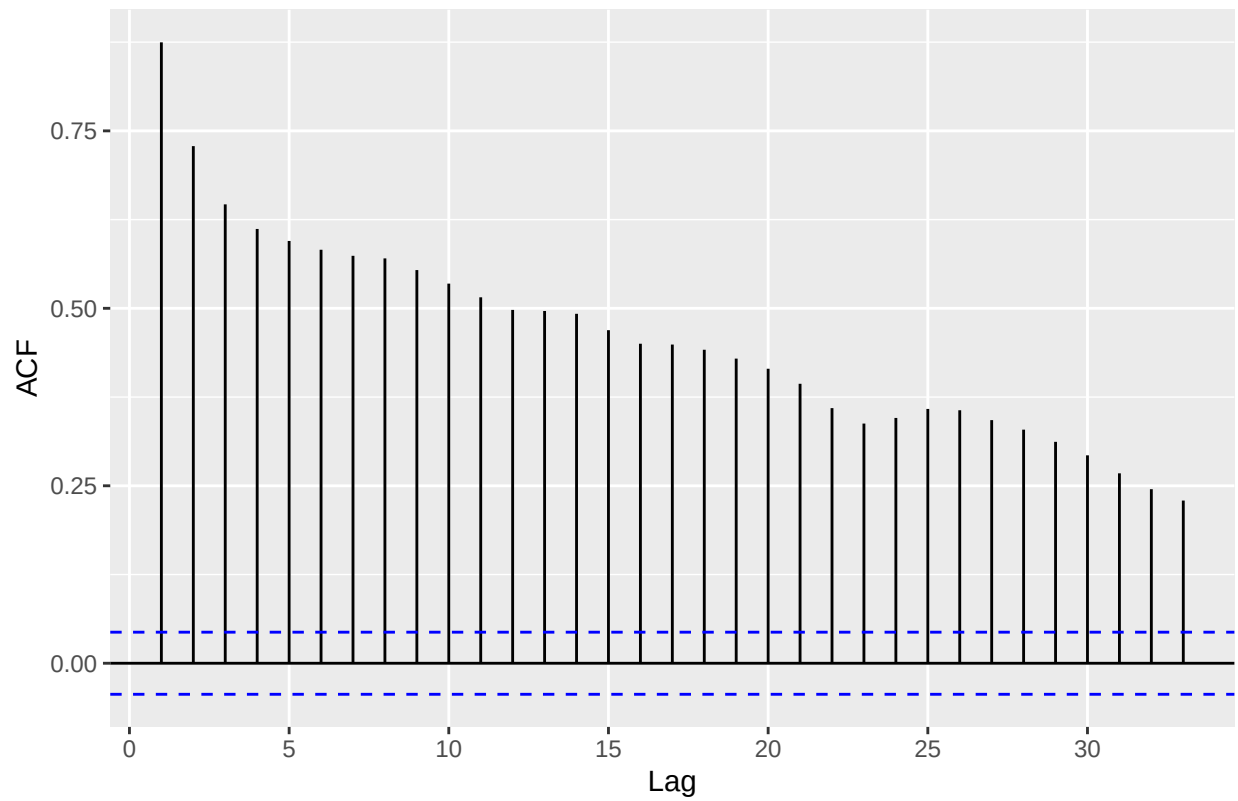
```

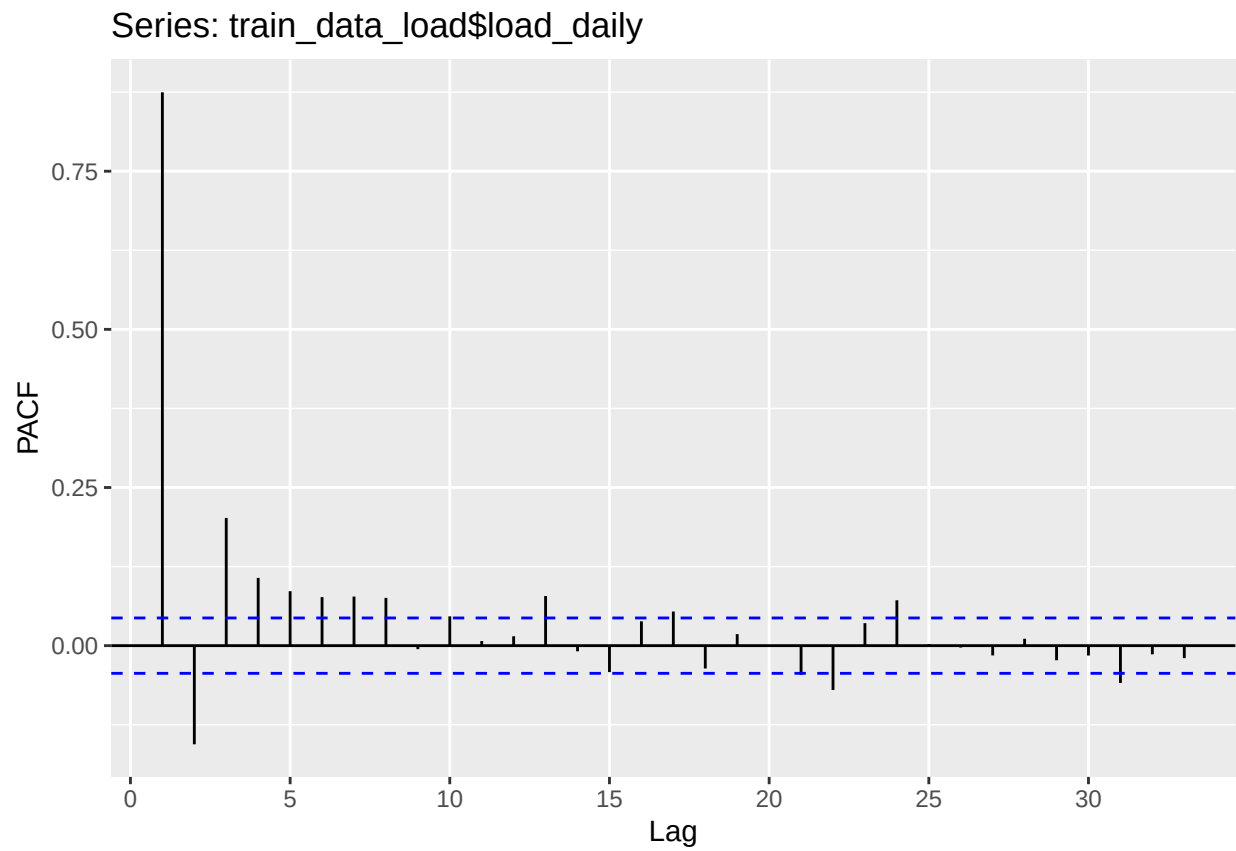
strong seasonality in weekly, yearly (reason for using msts)

```
ggAcf(train_data_load$load_daily)
```

Series: train_data_load\$load_daily

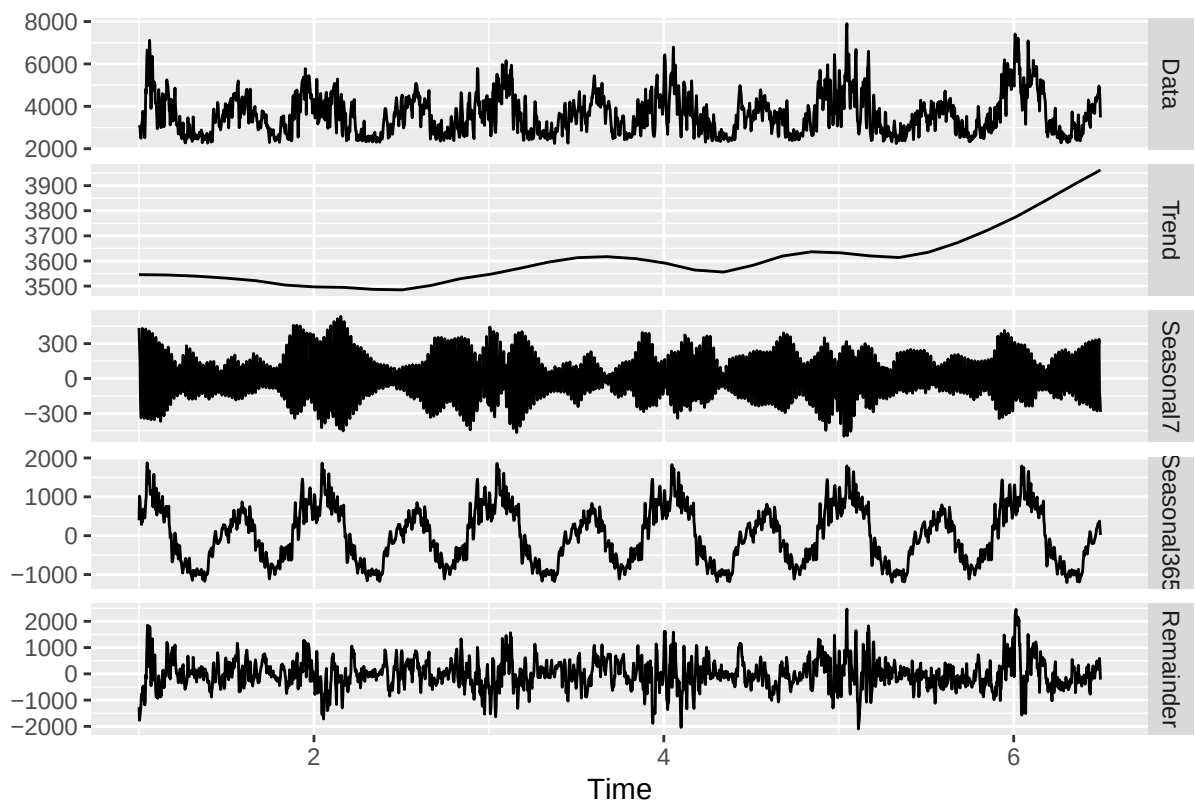


```
ggPacf(train_data_load$load_daily)
```



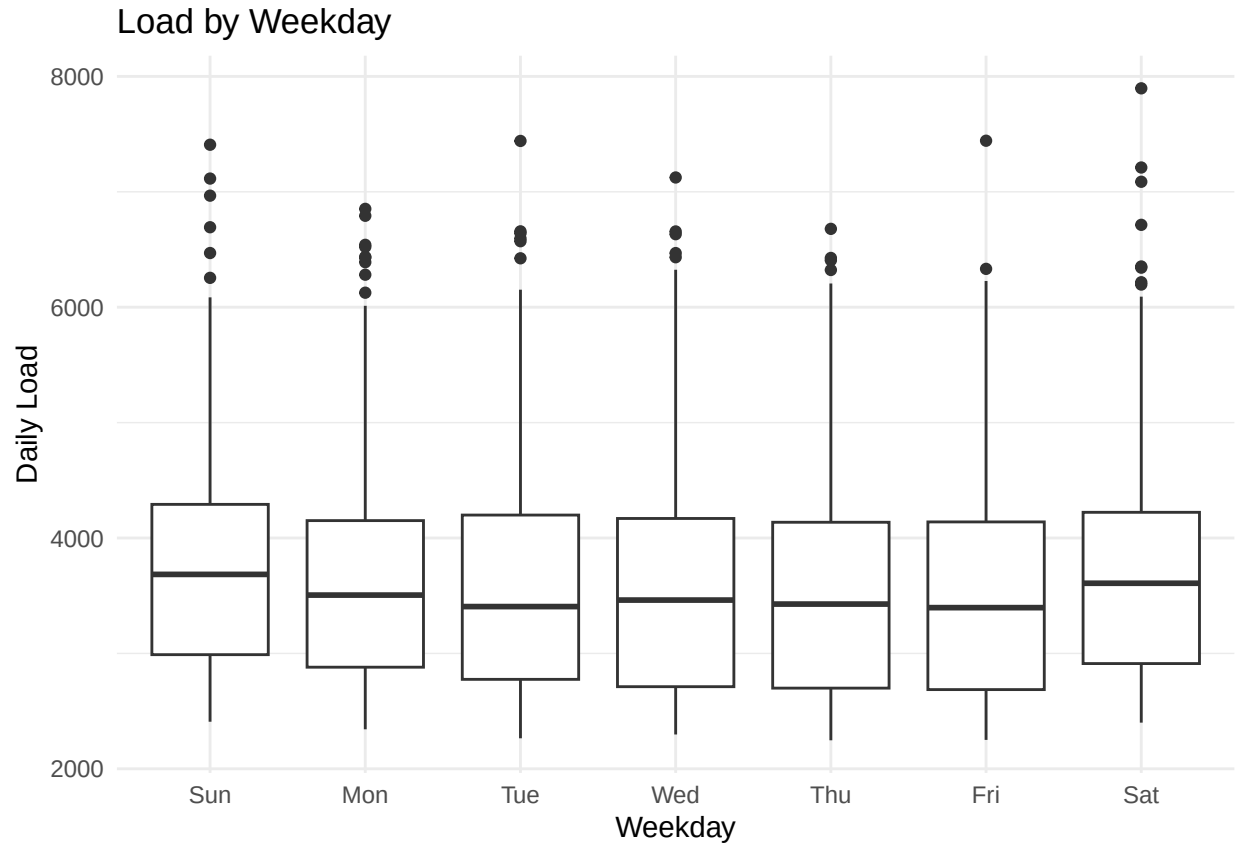
strong non-stationarity, low-order AR process

```
fit_mst1 <- mst1(train_msts_load)
autoplot(fit_mst1)
```



Trend exists(non-stationary), weak weekly seasonality, strong yearly seasonality

```
daily_load %>%
  mutate(weekday = wday(date, label = TRUE)) %>%
  ggplot(aes(x = weekday, y = load_daily)) +
  geom_boxplot() +
  labs(title = "Load by Weekday", x = "Weekday", y = "Daily Load") +
  theme_minimal()
```



Modeling

Among the models evaluated, the ETS model with mean bias correction achieved the best performance on Kaggle, with a score of 0.0928.

Ensemble models combining ETS and TBATS followed, with scores around 0.10, indicating that while ensembling provided some improvements, the bias-corrected ETS model remained the most effective.

For implementation, ETS was trained using a weekly seasonal structure (`ts` with frequency = 7), as it yielded comparable performance to multi-seasonal formulations. In contrast, TBATS was implemented using `msts` to capture multiple seasonal components in the data.

Model 1: ETS with Bias Correction

This model applies mean bias correction to the ETS forecast, which improves accuracy by correcting systematic underestimation observed in the validation set.

```
# ETS with Validation-based Bias Correction
# Refit ETS on training set to evaluate bias on validation
fit_ets_R <- ets(train_ts)
fc_ets_R <- forecast(fit_ets_R, h = nrow(valid_data))

# Compute mean systematic bias (actual - forecast) on validation
residuals_valid <- valid_data$load_daily - as.numeric(fc_ets_R$mean)
```

```

mean_bias <- mean(residuals_valid)
median_bias <- median(residuals_valid)
mean_fc <- mean(fc_ets_R$mean)

cat("Mean bias on validation:", round(mean_bias, 4), "\n")

## Mean bias on validation: 167.5916

cat("Median bias on validation:", round(median_bias, 4), "\n")

## Median bias on validation: 289.1033

cat("Mean forecast value:", round(mean_fc, 4), "\n")

## Mean forecast value: 4024.533

cat("Relative mean bias (%):", round(mean_bias / mean_fc * 100, 2), "%\n")

## Relative mean bias (%): 4.16 %

# Generate final forecast and apply mean bias correction
fit_ets_final <- ets(full_ts)
fc_ets_final <- forecast(fit_ets_final, h = nrow(sub_template))

# Apply additive mean bias correction
submission_ets_biasecorr <- sub_template %>%
  mutate(load = as.numeric(fc_ets_final$mean) + mean_bias)

write.csv(submission_ets_biasecorr, "output/submission_ets_biasecorr.csv", row.names = FALSE)

# Preview
head(submission_ets_biasecorr)

##           date      load
## 1 2011-07-01 3715.237
## 2 2011-07-02 3912.453
## 3 2011-07-03 3979.202
## 4 2011-07-04 3904.986
## 5 2011-07-05 3892.818
## 6 2011-07-06 3917.521

```

Model 2: ETS with Median Bias Correction

This model uses median bias correction, providing a more robust adjustment against extreme residual values compared to mean correction.

```
# ETS with Median Bias Correction

submission_ets_median_bias <- sub_template %>%
  mutate(load = as.numeric(fc_ets_final$mean) + median_bias)

write.csv(submission_ets_median_bias,
          "output/submission_ets_median_bias.csv", row.names = FALSE)

cat("Median bias used:", round(median_bias, 4), "\n")
```

```
## Median bias used: 289.1033
```

```
cat("Mean bias was: ", round(mean_bias, 4), "\n")
```

```
## Mean bias was:      167.5916
```

```
cat("Difference:      ", round(median_bias - mean_bias, 4), "\n")
```

```
## Difference:          121.5117
```

```
head(submission_ets_median_bias)
```

```
##      date      load
## 1 2011-07-01 3836.749
## 2 2011-07-02 4033.965
## 3 2011-07-03 4100.714
## 4 2011-07-04 4026.497
## 5 2011-07-05 4014.330
## 6 2011-07-06 4039.033
```

Model 3: Ensemble: 75% ETS + 25% TBATS

This ensemble combines 75% ETS and 25% TBATS forecasts, placing greater emphasis on ETS while incorporating multi-seasonal information from TBATS.

```
# Ensemble (75/25)

sub_ets <- read.csv("output/submission_ets.csv")
sub_tbats <- read.csv("output/submission_tbats.csv")

submission_ens_ets75_tbats25 <- sub_template %>%
  mutate(load = 0.75 * sub_ets$load + 0.25 * sub_tbats$load)

write.csv(submission_ens_ets75_tbats25,
          "output/submission_ens_ets75_tbats25.csv",
          row.names = FALSE)
```

Model 4: Ensemble: 70% ETS + 30% TBATS

This ensemble uses a 70/30 weighting between ETS and TBATS, slightly increasing the contribution of TBATS to better capture multiple seasonal patterns.

```
# ETS-heavy Weighted Ensemble
sub_ets    <- read.csv("output/submission_ets.csv")
sub_tbats  <- read.csv("output/submission_tbats.csv")

submission_ens_ets_heavy <- sub_template %>%
  mutate(load = 0.7 * sub_ets$load + 0.3 * sub_tbats$load)

write.csv(submission_ens_ets_heavy,
  "output/submission_ens_ets_heavy.csv", row.names = FALSE)
```

Model 5: Ensemble: 65% ETS + 35% TBATS

This ensemble further increases the TBATS contribution (65/35), allowing more flexibility in modeling complex seasonal structures.

```
# Ensemble (65/35)
submission_ens_ets65_tbats35 <- sub_template %>%
  mutate(load = 0.65 * sub_ets$load + 0.35 * sub_tbats$load)

write.csv(submission_ens_ets65_tbats35,
  "output/submission_ens_ets65_tbats35.csv",
  row.names = FALSE)
```

AI Usage

Generative AI tools (ChatGPT) were used to assist with code organization, debugging, and improving the clarity of written explanations.

Specifically, AI was used to refine R code structure, resolve errors related to file paths and LaTeX compilation, and enhance the presentation of results in a more professional format.

All modeling decisions, including model selection, bias correction methods, and evaluation strategies, were independently developed and verified by the authors.